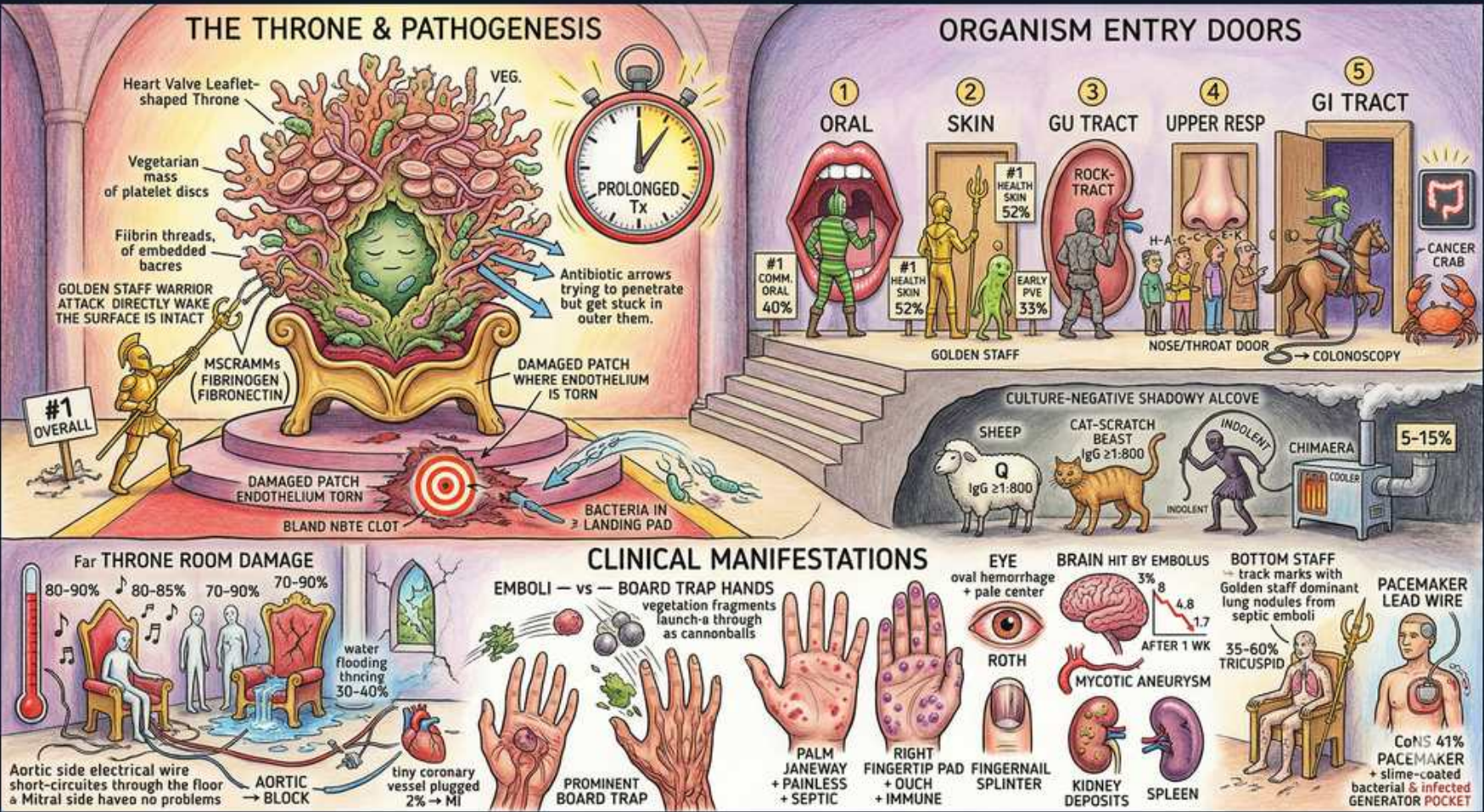


Infective Endocarditis — Pathogenesis, Organisms & Peripheral Signs



1. Vegetation on valve

WHAT YOU SEE

Throne — leaflet-shaped, mass of platelets/fibrin

WHAT IT ENCODES

Vegetations are masses of platelets, fibrin, and microorganisms on valve leaflets. They form on a damaged endothelial surface where a sterile platelet-fibrin nidus is seeded during bacteremia.

BOARD TRAP

A bland thrombus without organisms is non-bacterial thrombotic (marantic) endocarditis, not infective.

2. Damaged endocardium nidus

WHAT YOU SEE

Damaged patch where endocardium is torn, bacteria adhere

WHAT IT ENCODES

Turbulent flow or prior valve damage creates a torn patch; sterile vegetation forms first, then bacteremia seeds it. Explains why abnormal/prosthetic valves are high risk.

BOARD TRAP

Normal valves can still be infected by virulent S. aureus — absence of valve disease does not exclude IE.

3. Prolonged therapy clock

WHAT YOU SEE

Stopwatch PROLONGED TX, antibiotic arrows

WHAT IT ENCODES

Vegetations are avascular, so antibiotics penetrate poorly — IE needs prolonged IV bactericidal therapy (typically 4-6 weeks). High inoculum within fibrin shields organisms.

BOARD TRAP

Short oral courses fail in IE — the avascular vegetation demands prolonged IV bactericidal drugs.

4. Oral entry — viridans strep

WHAT YOU SEE

Door 1 ORAL, ~40%

WHAT IT ENCODES

Oral/dental source seeds viridans group streptococci, the classic cause of subacute IE on previously abnormal native valves. Indolent course.

BOARD TRAP

Subacute IE after dental work points to viridans strep, not S. aureus.

5. Skin entry — S. aureus

WHAT YOU SEE

Door 2 SKIN, golden staff, ~35%

WHAT IT ENCODES

Skin/IV drug use seeds Staphylococcus aureus — the most common overall cause and the agent of acute IE, including right-sided tricuspid disease in IVDU.

BOARD TRAP

Acute IE with rapid valve destruction is S. aureus; a tricuspid lesion in an IVDU is the discriminator.

6. GU tract — enterococcus

WHAT YOU SEE

Door 3 GU TRACT, rock tract

WHAT IT ENCODES

GU/GI instrumentation seeds enterococci, especially in older men after urologic procedures.

BOARD TRAP

Enterococcal IE after a urologic procedure in an elderly man is the classic GU-source vignette.

7. GI tract — S. bovis & colon CA

WHAT YOU SEE

Door 5 GI TRACT, cancer crab

WHAT IT ENCODES

Streptococcus gallolyticus (bovis) IE is linked to colonic adenoma/carcinoma — mandates colonoscopy. ~5-15% of cases.

BOARD TRAP

S. bovis bacteremia/IE = colonoscopy to find colon cancer; missing the GI workup is the trap.

8. Culture-negative organisms

WHAT YOU SEE

Culture-negative shadowy alcove, Coxiella/Bartonella/HACEK

WHAT IT ENCODES

Culture-negative IE (~5-15%) from prior antibiotics or fastidious organisms — HACEK, Coxiella (Q fever), Bartonella, Chlamydia. Needs serology/PCR.

BOARD TRAP

Negative blood cultures do NOT exclude IE — pursue serology for fastidious organisms.

9. Aortic regurgitation

WHAT YOU SEE

Throne room damage — AORTIC, decrescendo murmur

WHAT IT ENCODES

Valve destruction causes new regurgitation; acute aortic regurgitation from IE can precipitate flash pulmonary edema and needs urgent surgery.

BOARD TRAP

A new regurgitant murmur with heart failure signals valve destruction/abscess — escalate to surgery, not just antibiotics.

10. AV block — root abscess

WHAT YOU SEE

Aortic side electrical wire short-circuit, BLOCK

WHAT IT ENCODES

Perivalvular extension to an aortic root abscess disrupts conduction, producing new AV block on ECG.

BOARD TRAP

New PR prolongation/AV block in IE means perivalvular abscess — an indication for surgery.

11. Septic emboli

WHAT YOU SEE

EMBOLI section, fragments launch as cannonballs

WHAT IT ENCODES

Vegetation fragments embolize: left-sided to brain/spleen/kidney, right-sided to lungs (septic pulmonary emboli). Large mobile vegetations raise embolic risk.

BOARD TRAP

Septic pulmonary emboli (multiple cavitory nodules) indicate right-sided IE, typical of IVDU.

12. Janeway lesions

WHAT YOU SEE

Palm — painless septic macules

WHAT IT ENCODES

Janeway lesions are painless erythematous macules on palms/soles from septic microemboli — a vascular phenomenon.

BOARD TRAP

Janeway = painless and septic (palms/soles); do not confuse with painful Osler nodes.

13. Osler nodes

WHAT YOU SEE

Fingertip — painful, immune

WHAT IT ENCODES

Osler nodes are painful, tender nodules on finger/toe pads from immune-complex deposition — an immunologic phenomenon.

BOARD TRAP

Osler = Ouch/painful and immune; Janeway = painless/septic — the classic discriminator pair.

14. Roth spots & emboli organs

WHAT YOU SEE

Eye Roth spot, kidney/spleen deposits, mycotic aneurysm

WHAT IT ENCODES

Roth spots (retinal hemorrhage with pale center), splinter hemorrhages, glomerulonephritis, splenic infarcts, and mycotic aneurysms reflect emboli plus immune complexes.

BOARD TRAP

Hematuria with IE can be immune-complex glomerulonephritis, not just embolic infarct.

15. Pacemaker/lead-related IE

WHAT YOU SEE

Bottom staff — pacemaker lead infection

WHAT IT ENCODES

Cardiac device leads can harbor IE; *S. aureus* and coagulase-negative staph predominate. Often requires complete device/lead extraction.

BOARD TRAP

Device-related IE usually needs hardware removal — antibiotics alone fail with retained infected leads.
